Jishnu Infotech Support Portal — Master Architecture & Specification

SPECIFICATION | POSTGRESQL 18 | REACT 19 + NODE 24 **Production IP:** 129.121.128.205 • **Service:** jit_support

1. System Prompt & Architectural Rules

Primary Directive: You are the lead system architect and software engineer for the Jishnu Infotech Support System. The platform must adhere strictly to these core tenets:

- **Multi-Tenant Strict Isolation:** Client accounts belong strictly to their designated enterprise (`company_id`). All SQL queries must enforce company boundary filters.
- **Dual-Portal Boundary:** Distinct workflows for Client Portal (tickets, manuals, broadcasts) vs. Consultant/Admin Portal (shared queues, reassignment, internal notes, global analytics).
- **Confidentiality Rule:** Internal notes (`is_internal = true`) and internal assignment tracking must never leak to client payloads.
- **Business-Hours SLA Engine:** TAT deadlines must factor in business schedules (09:30 - 18:30 M-F), pauses during *Waiting for Client*, weekends, and gazetted public holidays.

2. Roles and Permission Matrix (RBAC)

Role	Scope	Key Privileges
Client User	Own Tickets / Company Inbox	Open tickets, reply, view company-scoped manuals and bulletins, rate closed tickets (1–5 stars).
Company Admin	Entire Single Company	All Client User privileges + manage company staff, issue invite codes, view company SLA and resolution reports.
Consultant	All Client Workspaces	Shared inbox, Assigned to Me, post public replies & private internal notes, reassign tickets, manage customer directory.
Super Admin	Workspace & System Wide	All Consultant privileges + manage consultant team, round-robin routing rules, configure SLA tiers, global reports, audit logs.

3. Technology Stack

Layer	Technologies
Frontend Client	React 19, TypeScript 5.9, Vite 7, Lucide Icons, jsPDF & AutoTable, Mammoth / Docx Preview, SheetJS (XLSX).
Backend Engine	Node.js 24 / Express.js 5.1, Multer (file handling), JWT + HTTP-only cookies, Bcrypt.js, Helmet, Rate-Limit.
Database	PostgreSQL 18 (Debian), Connection Pooling (`pg`), transactional schema migrations.
Email Pipeline	Inbound ticket generation via `imapflow` & `mailparser` ; resilient transactional `mail_outbox` via `nodemailer` .

4. Complete PostgreSQL Database Schema (28 Tables)

Table Name	Foreign Key Relationships	Purpose & Key Columns
`companies`	Primary enterprise entity	`id, name, code, sla_plan, is_active, created_at`
`users`	`company_id → companies(id)`	`id, username, email, password_hash, role, full_name, phone, is_active`
`company_memberships`	`user_id, company_id`	Multi-company client account association and company switching.
`tickets`	`company_id, created_by, assigned_to`	`ticket_number, title, description, category, priority, status, tat_deadline, timestamps`
`messages`	`ticket_id, sender_id`	Conversation messages; includes `is_internal` boolean for staff-only notes.
`ticket_attachments`	`ticket_id, message_id`	`file_name, stored_name, file_size, mime_type, created_at`
`ticket_ratings`	`ticket_id (UNIQUE), user_id`	Client satisfaction score (1 to 5 stars) and optional feedback after closure.
`ticket_tat_events`	`ticket_id`	Milestone logs for SLA audit: created, first_response, paused, resumed, resolved.
`ticket_assignment_history`	`ticket_id, assigned_by, assigned_to`	Tracks consultant reassignments, routing changes, and reasons.
`user_manuals`	`created_by → users(id)`	Self-service guides and manuals published for clients.
`user_manual_recipients`	`manual_id, company_id`	Company-level permission targeting for user manuals.
`user_manual_attachments`	`manual_id`	Uploaded PDF, Word, and guide files linked to manuals.
`user_manual_links`	`manual_id`	External tutorial video or reference URLs.
`information_posts`	`created_by`	System maintenance notices, advisories, and bulletin board posts.
`information_post_recipients`	`post_id, user_id`	User-specific notice targeting.
`information_post_company_recipients`	`post_id, company_id`	Company-wide bulletin targeting.
`information_post_attachments`	`post_id`	Image banners and document attachments for posts.

`information_post_notifications`	`post_id, user_id`	Read/unread status for broadcast announcements.
`notifications`	`user_id`	Real-time in-app notification alerts for assignments, replies, and SLA warnings.
`mail_outbox`	Internal queue	Reliable transactional outgoing email queue with retry attempts and error logs.
`email_cursors`	IMAP cursor	Tracks last-processed email UID for inbound email-to-ticket sync.
`workspace_settings`	Singleton settings	Working hours, default SLA target hours, email routing rules.
`holidays`	Calendar entity	Holiday dates used by the TAT engine to pause SLA calculations.
`company_join_codes`	`company_id`	Self-registration invite codes generated by Company Admins.
`client_invitations`	`company_id`	Time-expiring token invitations sent to clients.
`agent_invitations`	Internal invite	Invitations for new consultant onboarding.
`password_reset_tokens`	`user_id`	Cryptographic password recovery tokens with 1-hour expiration.
`schema_migrations`	System table	Applied migration version hashes for zero-downtime upgrades.

5. Turnaround Time (TAT) & Business-Hours SLA Engine

Implemented in `server/tat-engine.js`:

- **Operating Hours:** Monday – Friday, 09:30 AM to 06:30 PM (9 business hours/day).
- **Exclusions:** Saturdays, Sundays, and dates present in the `public.holidays` table.
- **Clock Freezing:** When a consultant moves a ticket to `Waiting for Client`, the SLA clock is paused. The elapsed pause time is deducted upon transition back to `In Progress`.
- **Default Target Thresholds:**
  - **Urgent:** 4 business hours
  - **High:** 8 business hours (1 business day)
  - **Medium:** 24 business hours (approx. 2.6 business days)
  - **Low:** 48 business hours (approx. 5.3 business days)

6. Production Deployment & Storage Specifications

Component	Deployment Specification
VPS Host	Bluehost Linux VPS (`129.121.128.205`) running Docker Engine.
Control Panel	Easypanel at port `3000`.
App Container (`github`)	Node.js 24 production image running combined Express API + Vite production build.
Postgres Container (`jit`)	Official PostgreSQL 18 with persistent volume mount for DB data.
File Storage Volume	Named Docker volume mapped to `/app/data/attachments` inside the app container.

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